Worksheet - 2

① Insurance companies collect accident data for drivers over two consecutive years. The probability that a randomly chosen driver is a 'good' driver (lower chance of being involved in an accident) is 'g' and probability that a randomly chosen driver is a 'bad' driver (higher chance of being involved in an accident) is "$1-g$".

Let G = event that a randomly chosen driver is a 'good' driver.

$$P(G) = g \quad ; \quad P(G^c) = 1-g$$

Moreover, a good driver has probability 'p_1' of an accident in any given year and a bad driver has probability 'p_2' of an accident in any given year.

So, let A = event that the randomly chosen driver has an accident in the 1st year.

& B = event that the randomly chosen driver has an accident in the 2nd year

Based on this information, answer the following:

(a) Express p_1 as a conditional probability using the labels provided.

(b) Express p_2 as a conditional probability using the labels provided.

(c) Express $P(B|G)$ and $P(B|G^c)$ in terms of p_1 and p_2.

(d) Assume that accidents in different years are independent once the driver's type is fixed. In other words, $P(A \cap B | G) = P(A|G)P(B|G)$

and $P(A \cap B | G^c) = P(A|G^c)P(B|G^c)$,

(i) Explain without computation whether A and B are independent events or not.

(ii) Find the probability that the driver does not meet with an accident in the 1st year, i.e. find $P(A^c)$.

(iii) Given that the driver does not have an accident in the first year, find the probability that the driver is a good driver.

find $P(G|A^c)$.

Solutions:

p_1 = probability that a good driver has an accident in a given year.

(a) $p_1 = P(A|G) = P(B|G)$

(b) p_2 = probability that a bad driver has an accident in a given year.

$p_2 = P(A|G^c) = P(B|G^c)$

(c) $p_1 = P(B|G)$; $p_2 = P(B|G^c)$

(d) (i) No. They are not independent.

Whether an accident happened in year 1 gives information about whether the driver is good or bad.

With that information, the probability of an accident in year 2 is affected

(ii) $P(A^c) = P(A^c|G)P(G) + P(A^c|G^c)P(G^c)$

$= (1 - P(A|G))P(G) + (1 - P(A|G^c))P(G^c)$

$= (1-p_1)g + (1-p_2)(1-g)$

(iii) $P(G|A^c) = \dfrac{P(A^c|G)P(G)}{P(A^c)}$

$$= \frac{(1-p_1)\,g}{(1-p_1)g + (1-p_2)(1-g)}$$